

Trainer Debrief — Lab 4

Voice of the Customer, Affinity and Kano — In Depth

Case study: Meridian Medical Devices — seal weld burst pressure

Objective: Capture and classify VOC for your capstone using affinity and Kano analysis (A1, K2).

***TRAINER ONLY.** Do not hand this out before the lab is complete.*

Expected analysis

lab-04-voc-affinity-kano.xlsx

The 40 verbatims cluster into roughly six affinity groups: product reliability, delivery performance, documentation and traceability, usability, service and support, commercial. 'No leaks' and 'documentation satisfies the HSA audit' are MUST-BE — you get zero credit for delivering them and a patient-safety event for missing them. 'OTIF 97%' and 'warranty settlement time' are ONE-DIMENSIONAL. 'Remote monitoring' and 'self-test on power-up' are ATTRACTIVE. Weight matters more than count: one procurement verbatim saying 'any leak is a patient safety event' outranks ten usability comments, and it is what points the project at seal weld integrity.

Every dataset is generated from a fixed seed, so these figures are reproducible — re-run the analysis on the shipped workbook and you will get these numbers.

What good looks like

You have fifteen or more verbatims, every verbatim sits in a named affinity cluster, every cluster carries a Kano classification, and your CTQ candidates trace to specific verbatims.

Where this sits in the capstone

A full re-derivation of VOC, affinity clustering and Kano classification — taught to the depth at which you could run this session yourself for a Green Belt team. Then apply all three to your own capstone process. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

The running process — figures that must reconcile

Checkpoint	Value	Established in
Baseline defect rate	2.9%	Lab 3
Measurement system	FAILS Gage R&R (~40% study variation)	Lab 11
Baseline capability	Pp 0.62 / Ppk 0.53	Lab 14
Proven Xs	clamp pressure, dwell time, electrode tip age	Labs 18-21
DOE optimum	clamp 47.4 bar, dwell 2.76 s	Labs 23-25
Improved capability	Ppk 1.40	Lab 27

Annual hard benefit	SGD 487,000 vs SGD 812,000 COPQ	Labs 1, 30
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If a learner's number disagrees with the column above, they have either used their own workplace data (fine — say so) or made an error (find it).

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